Vishnupraneeswar P

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.38
SRV Innovative Public School	Class XII (CBSE)	2023	95.00%
Rasi International School	Class X (CBSE)	2021	91.20%

EXPERIENCE

Applied Research Intern, Thomson Reuters Labs [*Labs & IIT Hyderbad verified report*]

- Trained a language model from scratch to investigate **latent recurrence** against Chain-of-Thought (CoT) reasoning.

- Engineered an end-to-end experimentation pipeline spanning distributed training and automated evaluation, optimized data pipeline to reduce GPU idle time from **~300s** to **~50s**.

- Designed compute-aware **ablation strategy** to isolate effects of architecture, data & training choices on model behavior.

- Identified **critical depth** for latent space collapse, recovered deep-model trainability through **carry-norm + pre-norm**.

- Redesigned token data storage to resolve distributed buffer overhead, reducing memory usage from **~440GB** to **~100GB**.

- Built an HS Classification system with multi-signal candidate extraction to handle sparsely described products.

- Built, deployed dedicated SME Review App to accelerate feedback-driven development with **on-demand** legal grounding.

- Exposed pipeline through modular stage-wise API endpoints with async response, reducing user-visible latency by **6 times**.

PROJECTS

Inverse Modeling of Fractional-Order Systems

- Built a fractional-order viscoelastic model to recover material parameters (,) using fractional PINNs.
- Replaced sequential finite differences with vectorized Caputo derivatives, yielding **20X** faster simulations.
- Found **Snake** activation excels in forward fitting but unstable for inverse tasks due to induced local minimas, hence adopted tanh + sine encoding for robust parameter recovery.
- Designed custom-grad operators for Caputo derivatives to identify bottlenecks in fractional computation.
- Validated robustness via , recovery under 10% white noise, confirming stable parameter determination.

Kernel Mapping in GemOS

- Designed user process virtual space layouts enabling valid representations of active memory mappings.
- Built lazy allocation to cut frame cloning on demand in fork-heavy workloads, exceeding eager allocation.
- Enhanced **CoW forking** by manipulating page table permissions and frame counts to optimize memory.

Lunar Elemental Mapping (ISRO-PS, Inter IIT Tech Meet)

- Preprocessed **2 TB** of Chandrayaan-II XRF data (5 yrs) in parallel, achieving 10X speedup over sequential baseline.
- Evaluated DBSCAN against SVM on **sparsely labeled lunar regions** to analyze regions with low solar activity.
- Selected **DBSCAN** for better performance in sparse spectral regimes and lower sensitivity to sparse labeling.
- Achieved **<5 km** resolution map (**Satellite: 25 km**) via **sub-pixel interpolation**, easing regional analysis.
- Deployed a web-based visualization tool featuring composition of elements on 2D, 3D lunar surfaces.

SKILLS

- **Programming Languages:** C, C++, Python
- **Libraries & Frameworks:** PyTorch, Scikit-Learn, NumPy, SciPy, Pandas, Matplotlib, Astropy
- **Engineering & Infra:** SLURM, AWS SageMaker, MLflow, Linux CLI, Bash
- **Scientific Tools:** PyMol, Quantum Espresso, Vesta, Xilinx, Solid Edge

COURSEWORK

- **AI and Scientific Computing:** Machine Learning, Distributed AI Training, Computer Aided Numerical Methods - 1, 2
- **Computer Science Foundations:** Data Structures and Algorithms, Operating Systems - 1, 2
- **Systems Modeling:** Intro to Hardware Description Languages, Modeling and Simulations, Atomistic Modeling
- **Math Foundations:** Calculus, Probability, Statistics, Transform Techniques

POSITIONS OF RESPONSIBILITY

Core Member, Epoch

- Organized community sessions to promote awareness and practical applications of AI among students.
- Contributed to machine learning projects focused on delivering positive outcomes to the community.

Undergrad Mentor, Sunshine

- Provided academic guidance to undergrads through regular mentoring, supporting strong performance.

Hospitality Coordinator, Milan

- Managed event planning, logistics and guest coordination for the General Championship of IIT Hyderabad.

EXTRACURRICULARS

Rural Development Center Volunteer

- Conducted educational sessions for underprivileged children from communities around IIT Hyderabad.
- Organized weekend science workshops to spark curiosity and promote STEM learning among rural students.

NSS Volunteer

- Contributed 121+ hours of service within a year (BTech requirement: 120 hours over 4 years).

SCHOLASTIC ACHIEVEMENTS

- **Wiz-Spell Bee:** National-level finalist among **80,000** candidates